

Measurement

Perimeter & Area · Composite Figures · Surface Area · Volume · Pyramids, Cones & Spheres · Optimization

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Find perimeter, area, surface area and volume of 2-D and 3-D figures, and reason about optimization.

Throughout this module use $\pi \approx 3.14$. Round answers involving π to **1 decimal place (1 d.p.)**. Every figure is described in words — read the dimensions carefully.

Warm-up (quick recall)

1. Find the area of a rectangle 6 cm long and 3 cm wide.
2. Find the perimeter of a square with side 5 cm.
3. State the formula for the area of a triangle.

Examples & Practice

For each skill, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Answers are on the last page.

Perimeter & Area of Basic Figures

Example:

A rectangle is 8 cm long and 5 cm wide. Find its perimeter and area.

Perimeter $P = 2(l + w) = 2(8 + 5) = 26$ cm.

Area $A = l \times w = 8 \times 5 = 40$ cm². Answer: $P = 26$ cm, $A = 40$ cm².

4. A square has side length 7 cm. Find its perimeter and area. **[L1]**

5. A triangle has base 10 cm and height 6 cm. Find its area. **[L1]**

6. A trapezoid has parallel sides 9 cm and 5 cm and a height of 4 cm. Find its area. **[L2]**

7. A circle has radius 5 cm. Using $\pi \approx 3.14$, find its circumference and area (1 d.p.). **[L3]**

Area of Composite Figures

Example:

A rectangle 10 cm by 6 cm has a semicircle of diameter 6 cm on one short end. Find the total area ($\pi \approx 3.14$, 1 d.p.).

Rectangle: $10 \times 6 = 60 \text{ cm}^2$. Semicircle radius = 3 cm: $\frac{1}{2} \times 3.14 \times 3^2 = 14.1 \text{ cm}^2$.

Total = $60 + 14.1 = 74.1 \text{ cm}^2$.

8. A figure is two rectangles joined along a 4 cm edge: one 5 cm by 4 cm and one 3 cm by 4 cm. Find the total area. **[L1]**

9. An L-shape is a 10 cm by 8 cm rectangle with a 4 cm by 3 cm rectangular corner removed. Find its area. **[L2]**

10. A square of side 8 cm has a quarter-circle of radius 8 cm removed from one corner. Find the remaining area ($\pi \approx 3.14$, 1 d.p.). **[L2]**

11. A track shape is a 20 cm by 10 cm rectangle with a semicircle of diameter 10 cm on each short end. Find the total area ($\pi \approx 3.14$, 1 d.p.). **[L3]**

Surface Area of Prisms & Cylinders

Example:

A rectangular prism is 6 cm by 4 cm by 3 cm. Find its surface area.
 $SA = 2(lw + lh + wh) = 2(6 \times 4 + 6 \times 3 + 4 \times 3) = 2(24 + 18 + 12)$.
 $= 2(54) = 108 \text{ cm}^2$.

12. A cube has edge length 5 cm. Find its surface area. **[L1]**
13. A cylinder has radius 3 cm and height 10 cm. Find its surface area ($SA = 2\pi r^2 + 2\pi rh$; $\pi \approx 3.14$, 1 d.p.). **[L2]**

14. A triangular prism is 12 cm long. Its triangular faces have base 6 cm and height 4 cm (equal sides 5 cm). Find the total surface area. **[L2]**
15. A closed cylinder has diameter 8 cm and height 15 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[L3]**

Volume of Prisms & Cylinders

Example:

A rectangular prism is 7 cm by 4 cm by 5 cm. Find its volume.

$$\begin{aligned} V &= \text{length} \times \text{width} \times \text{height} = 7 \times 4 \times 5. \\ &= 140 \text{ cm}^3. \end{aligned}$$

16. A cube has edge length 6 cm. Find its volume. **[L1]**

17. A cylinder has radius 5 cm and height 12 cm. Find its volume ($V = \pi r^2 h$; $\pi \approx 3.14$, 1 d.p.). **[L1]**

18. A triangular prism has a triangular base of area 15 cm^2 and length 9 cm. Find its volume. **[L2]**

19. A cylindrical tank has diameter 10 cm and height 14 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). **[L3]**

Surface Area & Volume of Pyramids, Cones & Spheres

Example:

A cone has radius 3 cm, height 4 cm and slant height 5 cm. Find its volume and surface area ($\pi \approx 3.14$, 1 d.p.).

$$V = (1/3)\pi r^2 h = (1/3)(3.14)(9)(4) = 37.7 \text{ cm}^3.$$

$$SA = \pi r^2 + \pi r l = 3.14(9) + 3.14(3)(5) = 28.3 + 47.1 = 75.4 \text{ cm}^2.$$

20. A sphere has radius 6 cm. Find its volume ($V = \frac{4}{3}\pi r^3$; $\pi \approx 3.14$, 1 d.p.). **[L1]**
21. A sphere has radius 4 cm. Find its surface area ($SA = 4\pi r^2$; $\pi \approx 3.14$, 1 d.p.). **[L2]**
22. A square-based pyramid has base edge 6 cm and height 10 cm. Find its volume ($V = \frac{1}{3} \times \text{base area} \times \text{height}$). **[L2]**
23. A cone has radius 6 cm and slant height 10 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[L3]**

2D & 3D Optimization

Example:

You have 40 m of fencing to enclose a rectangular garden. What dimensions give the maximum area?

For a fixed perimeter, a rectangle has maximum area when it is a square.

Side = $40 \div 4 = 10$ m, so $10 \text{ m} \times 10 \text{ m}$ gives the maximum area of 100 m^2 .

24. For a fixed perimeter, which rectangle encloses the greatest area? **[L1]**
25. A rectangular pen is built with 24 m of fencing to enclose the maximum area. Find its dimensions and that area. **[L2]**
26. For a fixed volume, which rectangular box uses the least material (smallest surface area)? **[L2]**

27. A closed box must hold 1000 cm^3 . What shape minimizes the surface area, and what are its dimensions? **[L3]**

Answers

Warm-up: W1) 18 cm^2 W2) 20 cm W3) $A = \frac{1}{2} \times \text{base} \times \text{height}$

4. (L1) $P = 4 \times 7 = 28 \text{ cm}$; $A = 7^2 = 49 \text{ cm}^2$.
5. (L1) $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$.
6. (L2) $A = \frac{1}{2}(a + b)h = \frac{1}{2}(9 + 5)(4) = 28 \text{ cm}^2$.
7. (L3) $C = 2 \times 3.14 \times 5 = 31.4 \text{ cm}$; $A = 3.14 \times 5^2 = 78.5 \text{ cm}^2$.
8. (L1) $5 \times 4 + 3 \times 4 = 20 + 12 = 32 \text{ cm}^2$.
9. (L2) $10 \times 8 - 4 \times 3 = 80 - 12 = 68 \text{ cm}^2$.
10. (L2) $64 - \frac{1}{4} \times 3.14 \times 8^2 = 64 - 50.2 = 13.8 \text{ cm}^2$.
11. (L3) Rectangle $20 \times 10 = 200$; two semicircles = one circle $r = 5$: $3.14 \times 5^2 = 78.5$.
Total = 278.5 cm^2 .
12. (L1) $SA = 6 \times \text{edge}^2 = 6 \times 5^2 = 150 \text{ cm}^2$.
13. (L2) $2 \times 3.14 \times 3^2 + 2 \times 3.14 \times 3 \times 10 = 56.5 + 188.4 = 244.9 \text{ cm}^2$.
14. (L2) Triangles: $2 \times (\frac{1}{2} \times 6 \times 4) = 24$; rectangles: $(6 + 5 + 5) \times 12 = 192$; $SA = 216 \text{ cm}^2$.
15. (L3) $r = 4$: $2 \times 3.14 \times 4^2 + 2 \times 3.14 \times 4 \times 15 = 100.5 + 376.8 = 477.3 \text{ cm}^2$.
16. (L1) $V = \text{edge}^3 = 6^3 = 216 \text{ cm}^3$.
17. (L1) $V = 3.14 \times 5^2 \times 12 = 942.0 \text{ cm}^3$.
18. (L2) $V = \text{base area} \times \text{length} = 15 \times 9 = 135 \text{ cm}^3$.
19. (L3) $r = 5$: $V = 3.14 \times 5^2 \times 14 = 1099.0 \text{ cm}^3$.
20. (L1) $V = (\frac{4}{3}) \times 3.14 \times 6^3 = 904.3 \text{ cm}^3$.
21. (L2) $SA = 4 \times 3.14 \times 4^2 = 201.0 \text{ cm}^2$.
22. (L2) $V = (\frac{1}{3}) \times 6^2 \times 10 = (\frac{1}{3}) \times 36 \times 10 = 120 \text{ cm}^3$.
23. (L3) $SA = 3.14 \times 6^2 + 3.14 \times 6 \times 10 = 113.0 + 188.4 = 301.4 \text{ cm}^2$.
24. (L1) A square (all four sides equal).
25. (L2) Square: side = $24 \div 4 = 6 \text{ m}$; area = $6 \times 6 = 36 \text{ m}^2$.
26. (L2) A cube (all edges equal).
27. (L3) A cube; edge = $\sqrt[3]{1000} = 10 \text{ cm}$, so $10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$.